PRD

FACTS

File & Case Tracking System

A comprehensive case management and welfare tracking platform for Baaseteen & SHND organizations

VERSION	DATE	STATUS
3.7	April 23, 2025	Final

• LIVE & OPERATIONAL

Prepared by FACTS Development Team · Classification: Internal

Product Requirements Document (PRD)

FACTS — File & Case Tracking System

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Document Control

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1. Product Vision & Overview

1.1 Vision Statement

FACTS aims to be the definitive digital platform for welfare case management — eliminating fragmented paper-based processes, enforcing consistent multi-level approval workflows, and providing every stakeholder with real-time visibility from the moment a case is identified through to financial disbursement and fund utilization tracking.

1.2 Product Summary

FACTS is a full-stack web application built for Baaseteen and SHND organizations. It digitizes the entire welfare case lifecycle: from eligibility screening and applicant profiling through counseling, multi-authority approvals, financial disbursement scheduling, and ongoing quarterly reassessment. The system enforces role-scoped access, automated notifications, SLA tracking, and comprehensive audit trails at every step.

Key Capabilities at a Glance:

CAPABILITY	WHAT IT DOES
Case Identification	Pre-assessment eligibility screening with ITS integration, income evaluation, and auto-case creation on approval
Case Management	Full lifecycle tracking with configurable multi-stage approval workflows, comments, and audit trails
Applicant Profiles	Rich applicant records auto-populated from the ITS API including photos, family details, and contact info
Counseling Forms	Multi-section digital forms (7 sections) replacing paper-based counseling documentation
Cover Letters	Automated PDF generation for case cover letters with submission workflows
Payment Schedules	Yearly disbursement plans with monthly repayment tracking and confirmation
Fund Utilization	Quarterly reassessment reports with configurable dynamic form builder and photo documentation
Welfare Checklist	Configurable checklist categories and items with per-case completion tracking
RBAC	Granular role-based access at resource + action level with Jamiat/Jamaat organizational scoping
SLA Tracking	Per-stage SLA values with automated breach detection and in-app alerts
Notifications	In-app alerts for status changes, assignments, and SLA warnings
Analytics Dashboard	Role-specific dashboards with case pipeline views and activity feeds

1.3 Product Positioning

ATTRIBUTE	DETAIL
For	Baaseteen and SHND welfare organizations
Who	Administrators, DCMs, Counselors, Welfare Reviewers, ZI Officers, Executive Approvers, Finance staff
FACTS is	A web-based case management and welfare tracking system
That	Centralizes every step of the welfare case lifecycle from screening to fund monitoring
Unlike	Manual paper registers, disconnected spreadsheets, and ad-hoc email approvals
Our Product	Provides a unified, role-scoped platform with configurable workflows, automated PDF generation, real-time notifications, and SLA enforcement

2. Problem Statement

2.1 Current Challenges

CHALLENGE	IMPACT
Paper-based case intake	Eligibility assessments were manual; no central record of who was screened, their income, or their outcome
Disconnected approval chain	Cases moved between counselors, welfare reviewers, ZI, and executives via email and in-person; no trackable audit trail
No standardized counseling forms	Each counselor used their own format; comparison across cases was impossible
Manual financial tracking	Payment disbursements and repayment schedules were tracked in spreadsheets susceptible to errors and loss
No fund utilization visibility	There was no systematic way to track how disbursed funds were used quarter-by-quarter
Organizational scope confusion	Staff in one Jamiat could view or act on cases belonging to another, violating data boundaries
Duplicate case risk	No enforcement preventing an applicant from having multiple concurrent active cases
Zero SLA enforcement	Cases could sit at a stage indefinitely with no system-level warnings or escalation

2.2 Desired Outcome

A single platform where:

- **Field workers** can screen applicants and submit eligibility assessments digitally with ITS auto-fill
- **Counselors** can complete structured multi-section forms and attach supporting documents
- **Welfare reviewers, ZI officers, and executives** can approve or return cases through a traceable workflow
- **Finance staff** can manage payment schedules and confirm disbursements
- **Administrators** can configure workflows, roles, master data, and monitor every case from a single dashboard
- **All users** receive real-time notifications and see only the data relevant to their organizational scope

3. Goals & Success Metrics

3.1 Product Goals

ID	GOAL	PRIORITY
G-1	Digitize the full welfare case lifecycle end-to-end	Critical
G-2	Enforce configurable multi-level approval workflows with audit trails	Critical
G-3	Ensure organizational data scoping (Jamiat/Jamaat isolation)	Critical
G-4	Prevent duplicate active cases per ITS number	High
G-5	Provide real-time SLA tracking and breach alerts	High
G-6	Automate PDF generation for cover letters and quarterly reports	High
G-7	Enable quarterly fund utilization tracking per disbursed case	High
G-8	Deliver role-specific dashboards with actionable analytics	Medium
G-9	Support bulk data import/export via Excel for all master entities	Medium

3.2 Key Performance Indicators (KPIs)

KPI	TARGET	MEASUREMENT METHOD
System uptime	≥ 99.5%	Server monitoring
Case lifecycle fully digital	100% of new cases	Database case records
Average page load time	< 3 seconds	API response monitoring
Duplicate case violations	0 per month	One-active-case-per-ITS enforcement log
SLA breach notifications sent	100% of breached stages	Notification records
Counseling form digital completion	100% of assigned cases	Form completion records
ITS auto-fill success rate	≥ 90%	API fetch logs
Report generation time	< 10 seconds	PDF generation timing

3.3 Success Criteria

- All case identifications, approvals, and closures occur within the platform — zero paper-based processing
- Every case progresses through traceable workflow stages with timestamped comments
- Organizational scoping prevents any user from viewing out-of-scope Jamiat/Jamaat data
- Payment schedules and disbursements are tracked in-system for every approved case
- Quarterly fund utilization reports are submitted and stored digitally for all active disbursed cases
- Administrators can configure workflow stages, roles, and master data without code changes

4. Target Users & Personas

4.1 Administrator / Super Admin

ATTRIBUTE	DETAIL
Role	System Administrator
Goals	Configure and oversee all system operations; manage users, roles, master data, and workflows
Key Tasks	User CRUD, role/permission management, workflow stage configuration, master data management, dashboard oversight
Technical Skill	High
Access Level	Full system access — no Jamiat/Jamaat scope restrictions

4.2 DCM — Deputy Case Manager

ATTRIBUTE	DETAIL
Role	Case management lead
Goals	Create and assign cases, oversee counselors, submit cases to welfare review
Key Tasks	Case creation, counselor assignment, cover letter submission, case resubmission after rework
Technical Skill	Moderate
Access Level	Jamiat-scoped — sees cases in assigned Jamiats only

4.3 Counselor

ATTRIBUTE	DETAIL
Role	Welfare counseling practitioner
Goals	Complete detailed multi-section counseling forms for assigned cases
Key Tasks	Fill counseling form sections, capture digital signatures, upload attachments, mark forms complete
Technical Skill	Low to Moderate
Access Level	Jamaat-scoped — sees cases in assigned Jamaats only

4.4 Welfare Reviewer

ATTRIBUTE	DETAIL
Role	Welfare department reviewer
Goals	Review counseling quality, complete welfare checklist, approve or send cases for rework
Key Tasks	Checklist review, welfare approve/reject/rework, forward to ZI
Technical Skill	Moderate
Access Level	Jamiat-scoped

4.5 ZI — Zonal Incharge

ATTRIBUTE	DETAIL
Role	Zonal-level approver
Goals	Review welfare-approved cases and provide approval or rejection

ATTRIBUTE	DETAIL
Key Tasks	ZI approve/reject with comments, forwarding to executive levels
Technical Skill	Low to Moderate
Access Level	Scoped to assigned zones

4.6 Executive Management

ATTRIBUTE	DETAIL
Role	Senior organizational leaders (up to 5 levels)
Goals	Final approval or rework decisions at their assigned executive level
Key Tasks	Executive approve/rework, view case summaries and cover letters
Technical Skill	Low
Access Level	Read-only on case data; approval actions scoped to their level

4.7 Finance Staff

ATTRIBUTE	DETAIL
Role	Financial operations team
Goals	Manage payment schedules and confirm actual disbursements
Key Tasks	Create/edit payment schedules, confirm disbursement with dates, track repayments
Technical Skill	Moderate
Access Level	Payment management module; case read access

4.8 User Distribution

ROLE	ESTIMATED COUNT	ACCESS SCOPE
Super Admin / Admin	2–5	Full system access
DCM	10–20	Assigned Jamiats
Counselor	20–50	Assigned Jamaats
Welfare Reviewer	5–15	Assigned Jamiats
ZI / Executive	5–15	Zonal / Executive level
Finance	3–10	Payment management

5. User Stories

5.1 Authentication & Access

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-A-001	Any User	Log in with username/email and password	I can securely access the system	Critical
US-A-002	Any User	See a role-specific dashboard after login	I immediately see relevant cases and actions	Critical
US-A-003	Any User	Log out and have my session destroyed	My account is protected on shared devices	Critical
US-A-004	Admin	Register new users with role assignments	Staff can access the system with the right permissions	Critical

5.2 Case Identification

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-CI-001	Field Worker	Enter an applicant's ITS number and auto-fill their details	I don't manually type data already in the system	Critical
US-CI-002	Field Worker	Record income, family count, and assign a case type	Eligibility can be assessed consistently	Critical
US-CI-003	Reviewer	Mark a screening as eligible or ineligible with remarks	Eligible ones auto-create a case for the DCM	Critical
US-CI-004	Reviewer	Edit income assessment fields on pending/ineligible records	Errors can be corrected without creating duplicates	High
US-CI-005	System	Block duplicate ITS numbers from being screened twice	One active case per applicant is enforced	Critical

5.3 Case Management

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-CM-001	DCM	Create and assign cases to counselors	Counseling begins promptly	Critical
US-CM-002	Welfare Reviewer	Approve or return cases for rework	Only quality cases advance to ZI review	Critical
US-CM-003	ZI / Executive	Approve or reject cases at my level	The case moves to the next stage or is returned	Critical
US-CM-004	Any User	Add comments at any workflow stage	Context and reasoning are preserved in the audit trail	High
US-CM-005	Admin	Close a case with a reason and supporting documents	Case closure is formally recorded	High
US-CM-006	Any User	See SLA status (on time / at risk / breached) for each case	Delayed cases can be escalated before they breach	High

5.4 Counseling & Cover Letters

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-CF-001	Counselor	Fill multi-section counseling forms section by section	I can save progress without losing data	Critical

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-CF-002	Counselor	Capture digital signatures within the declaration section	No physical paper signing is needed	High
US-CF-003	DCM	Generate a cover letter PDF with one click	Formatted cover letters are ready for submission	High
US-CF-004	Welfare Reviewer	Sign the cover letter form at the welfare stage	My approval is formally recorded	High

5.5 Financial & Fund Utilization

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-FIN-001	Finance	Create a payment schedule with yearly disbursement amounts	The repayment plan is clear from the start	Critical
US-FIN-002	Finance	Confirm a disbursement with the actual date	Records reflect real payment activity	Critical
US-FIN-003	DCM / Counselor	Submit quarterly fund utilization reports with photos	The organization tracks how funds are being used	High
US-FIN-004	Admin	Export fund utilization reports as PDF	Reports can be shared with leadership offline	Medium

5.6 Administration

ID	AS A...	I WANT TO...	SO THAT...	PRIORITY
US-ADM-001	Admin	Configure workflow stages per case type with role assignments	Approval pipelines match each program's needs	Critical
US-ADM-002	Admin	Import users and Jamiat/Jamaat data from Excel	Large datasets can be onboarded efficiently	High
US-ADM-003	Admin	Set SLA values per workflow stage	Automated breach alerts are calibrated correctly	High
US-ADM-004	Admin	Manage the fund utilization form builder	Quarterly report questions can evolve without developer changes	Medium

6. Feature Requirements

6.1 Authentication & Access Control

Purpose: Securely authenticate users and enforce role-based, organizationally-scoped access across the platform.

FEATURE	DESCRIPTION	PRIORITY
JWT authentication	Stateless JWT tokens (24-hour expiry) issued on login; verified on every request	Critical
bcrypt password storage	All passwords hashed with bcryptjs; minimum 6 characters enforced on change	Critical
Multi-identifier login	Users can authenticate with either username or email address	High
Role-based access control	Permissions defined at resource + action level (e.g., `cases:create`, `case_identification:approve`)	Critical
Organizational scope enforcement	Jamiat-scope users see only Jamiat-assigned cases; Jamaat-scope users see only Jamaat-assigned cases	Critical
Multi-role assignment	Users can hold multiple roles simultaneously; primary display role can be toggled	High
Token user-status check	Every authenticated request checks whether the user account is still active	High
Rate limiting on login	Brute-force protection on the login endpoint via `express-rate-limit`	Medium

6.2 User & Role Management

Purpose: Allow administrators to create, configure, and manage all user accounts and permission roles.

FEATURE	DESCRIPTION	PRIORITY
User CRUD	Create/read/update/delete users with name, email, phone, ITS number, photo (base64), and role assignments	Critical
Role CRUD	Create/edit roles with display name, description, and granular permission sets	Critical
Permission assignment	15 resources × multiple actions; assigned at role level and enforced by middleware	Critical
Counseling form stage permissions	Per-role read/update access configurable per counseling form stage	High
Executive level assignment	Assign users to numbered executive approval levels (1–5); Admin/Super Admin only	High
Activate/deactivate users	Toggle user status; deactivated users cannot authenticate	High
Bulk import via Excel	Import users from .xlsx/.xls with per-row validation and error reporting	Medium
ITS API sync	Fetch and update contact details from the external ITS system for users with ITS numbers	Medium
Export to Excel	Download user lists filtered by role or status	Medium

6.3 Case Identification

Purpose: Pre-assess applicant eligibility before creating a full case, ensuring only eligible candidates enter the workflow.

FEATURE	DESCRIPTION	PRIORITY
ITS auto-fill	Enter ITS number → system fetches name, age, gender, photo, contact, Jamiat/Jamaat from DB or external API	Critical
Duplicate ITS detection	System checks if ITS number already has an existing case identification record and blocks re-entry	Critical
Income assessment	Record individual income, family income, total family members, earning members, and case type assignment	Critical

FEATURE	DESCRIPTION	PRIORITY
Eligible / Ineligible review	Authorized reviewers mark records as eligible (auto-creates a case) or ineligible with remarks	Critical
Edit assessment fields	Edit income fields and eligible_in on pending or ineligible records; changes logged to audit table	High
Revert to pending	Revert an ineligible record back to pending for re-review	High
Income change audit log	Every income field modification is recorded with old value, new value, changed_by, and timestamp	High
Filter and search	Filter by status, case type, Jamiat, Jamaat; search by ITS or full name with pagination	High

6.4 Case Management

Purpose: Manage the full lifecycle of welfare cases from creation to closure through configurable multi-stage workflows.

FEATURE	DESCRIPTION	PRIORITY
Case creation	Create cases manually or auto-generate from eligible case identifications; assigns auto-incremented case number (e.g., BS-0001)	Critical
Case assignment	Assign DCM user and counselor per case; restrict visible cases by Jamiat/Jamaat scope	Critical
Workflow actions	Approve, reject, rework, resubmit — each action advances or returns the case to a defined stage	Critical
Multi-level executive approval	Cases route through up to 5 configurable executive levels; next level determined automatically	Critical
Stage-specific comments	Users add general or workflow-step comments; all comments time-stamped and stored per case	High
Status history	Every status transition is logged with from/to status, changed_by, and timestamp	High
SLA tracking	Each stage has configurable SLA value and warning threshold; system calculates on_time / at_risk / breached	High
Case closure	Close a case with a reason and 1–10 supporting documents (PDF, DOC, DOCX, XLS, XLSX, JPG, PNG, max 10 MB each)	High
Duplicate case flag	Cases are flagged if the same ITS number has multiple active cases	Medium
Filter and search	Filter by status, case type, assigned DCM, counselor, Jamiat, Jamaat, workflow stage, duplicate flag	High

6.5 Applicant Management

Purpose: Maintain rich applicant profiles that serve as the foundation for case counseling and financial planning.

FEATURE	DESCRIPTION	PRIORITY
Applicant profiles	Full profiles: personal details, contact, photo (base64), address, Jamiat/Jamaat affiliation	Critical
ITS auto-fetch	Fetch applicant data from external ITS API by ITS number; fallback to local DB if already stored	Critical
Bulk import from Excel	Import multiple ITS numbers at once; system fetches details from ITS API for each	High
Bulk fetch from API	Trigger batch API fetch for all pending applicants without details	Medium
One-active-case enforcement	System checks for existing active cases when creating new ones; blocks creation if found	Critical
Export template	Download pre-formatted Excel template for ITS number bulk import	Low

6.6 Counseling Forms

Purpose: Replace paper-based counseling documentation with a structured, section-by-section digital form linked to each case.

6.6.1 Form Sections

#	SECTION	KEY CONTENT
1	Personal Details	ITS, name, age, education, Jamiat/Jamaat, contact, address, occupation
2	Family Details	Family structure, wellbeing areas (food/housing/health/education/deeni), income & expense breakdown, assets & liabilities, family members table
3	Assessment	Applicant background, proposed business details, counselor evaluation of demand/supply/competition
4	Financial Assistance	Assistance required, self-funding, repayment schedule, QH fields, support mentors
5	Economic Growth	5-year revenue, expense, and profit projections; asset/liability trajectory
6	Declaration	Applicant, counselor, and TR Committee digital signatures with date and confirmation text
7	Attachments	Workplace photos, quotations, income tax returns, PAN card, Aadhar card, other documents

6.6.2 Counseling Form Features

FEATURE	DESCRIPTION	PRIORITY
Section-by-section save	Each section saves independently via a PUT endpoint — partial progress is never lost	Critical
Digital signatures	Signatures captured by upload or drawing pad; stored as base64 or file path	High
Auto-populated fields	Applicant info, counselor info, and financial data pre-filled from linked case data	High
Family members table	Dynamic table for adding/removing family members with relation, education, occupation, and income	High
Form completion tracking	System tracks which sections are complete; marks overall form complete when all sections done	High
Stage-level permissions	Role-specific read/update access configurable per section via `counseling_form_stage_permissions`	High

6.7 Cover Letters

Purpose: Automate generation of formatted cover letters that summarize the case for executive review.

FEATURE	DESCRIPTION	PRIORITY
Cover letter form	Structured form with applicant summary, counselor info, financial figures, projected income, and assistance amounts	High
Multi-party signatures	Welfare department, ZI, and Operations Head signatures with date fields (upload or draw)	High
PDF generation	Generate cover letter as PDF; stored and linked to the case	High
Submit workflow	Submitting the form advances the case; only authorized roles can submit	High
Download	Any authorized user can download the generated PDF	High

6.8 Payment Management

Purpose: Track all financial disbursements and repayment plans for approved cases.

FEATURE	DESCRIPTION	PRIORITY
Payment schedule creation	Define up to 5 yearly disbursements with disbursement year, month, and amount; supports Qardan Hasana and Enayat (grant) types	Critical
Repayment schedule	Auto-generate monthly repayment entries for Qardan Hasana based on months and start date	High
Disbursement confirmation	Finance staff confirm actual disbursement with date; records disbursed_by and disbursed_at	Critical
Schedule editing	Edit schedules before disbursement; locked after confirmation	Medium

6.9 Fund Utilization

Purpose: Track how disbursed funds are utilized quarter-by-quarter through configurable structured reports.

FEATURE	DESCRIPTION	PRIORITY
Quarterly reports	Create/edit/delete fund utilization reports per case; one report per quarter per year	High
Dynamic form builder	Admins define sections and questions (text, number, select, file upload, table types) without code changes	High
Auto-fill fields	Known values (e.g., Mauze from applicant data) auto-populate into matching form fields	Medium
Photo documentation	Upload photos per report categorized as: shop_site, stock_equipment, applicant_work, other	High
File attachments per question	Upload supporting files linked to specific form questions	Medium
PDF report generation	Generate PDF quarterly report via PDFKit or Puppeteer HTML template	High
Section/question reordering	Drag-and-drop reorder for form sections and questions with sort_order persistence	Medium

6.10 Welfare Checklist

Purpose: Provide structured quality checklists for welfare reviewers to assess counseling form completeness.

FEATURE	DESCRIPTION	PRIORITY
Category management	Create/edit/delete checklist categories with sort order; drag-and-drop reordering	Medium
Item management	Define checklist items per category with form_section reference, compulsory flag, and description	Medium
Per-case responses	Submit YES/NO/NA responses per item with comments; overall remarks field	High
Completion status	Track total vs filled items; calculate completion percentage per case	High

6.11 Workflow Management

Purpose: Enable administrators to configure approval pipelines without code changes.

FEATURE	DESCRIPTION	PRIORITY
Stage CRUD	Create/edit/soft-delete/restore workflow stages with name, key, sort order, and case type association	Critical
Role assignment to stages	Assign roles to stages with per-role permission flags: can_approve, can_reject, can_review, can_view, can_edit, can_delete, can_create_case, can_fill_case	Critical
User assignment to stages	Assign individual users to stages with same permission flags	High
SLA configuration	Set SLA value + unit (hours/days/business_days/weeks/months) and warning threshold per stage	High

FEATURE	DESCRIPTION	PRIORITY
Status mappings	Define which case statuses correspond to each workflow stage	High
Stage reordering	Drag-and-drop reorder stages with persisted sort_order	Medium
Auto-advance on approve	Configurable flag to automatically advance to next stage on approval action	Medium

6.12 Notification System

Purpose: Keep all users informed about case changes, assignments, and SLA alerts in real time.

TRIGGER EVENT	RECIPIENTS
Case status changed	Assigned DCM and counselor
Case assigned to counselor	Assigned counselor
Workflow action taken (approve/reject)	Relevant stage participants
SLA breach / at-risk	Responsible role users

FEATURE	DESCRIPTION	PRIORITY
Notification list	View all notifications with title, message, type, case reference, and date	High
Unread count badge	Live count in navigation bar	High
Mark as read	Individual and bulk mark-as-read	Medium
SLA check trigger	Admin-triggered SLA calculation via `POST /api/notifications/check-sla`	Medium

6.13 Reporting & Analytics

Purpose: Provide role-appropriate dashboards and exportable reports for decision-making.

REPORT	DESCRIPTION	AUDIENCE
Overview Dashboard	Total cases by status (draft, counseling, welfare review, ZI, executive, finance) with pipeline counts	All roles (filtered)
Recent Activities	Latest status changes with case number, applicant name, and who changed it	All roles
Case Pipeline	Case counts grouped by current status for visual pipeline representation	All roles
User Statistics	Total/active/inactive users broken down by role	Admin, DCM, ZI
Quarterly Report PDF	Full fund utilization quarterly report as downloadable PDF	Admin, Finance, DCM
Excel Export	Export users, Jamiat/Jamaat master data from the system	Admin

6.14 Attachment Management

Purpose: Manage file uploads across all case stages with type validation and size limits.

FEATURE	DESCRIPTION	PRIORITY
Single file upload	Upload one file per request; linked to a case with optional stage tag	High
Multi-file upload	Upload up to 10 files in one request	High
File type validation	Allowed: PDF, DOC, DOCX, XLS, XLSX, JPG, JPEG, PNG; enforced by Multer middleware	Critical
File size limit	Maximum 10 MB per file	Critical
Stage-tagged uploads	Attachments can be tagged with the workflow stage at which they were uploaded	Medium
Download	Authorized users download attachments via secure endpoint	High

FEATURE	DESCRIPTION	PRIORITY
Attachment statistics	Total file count, total size, and per-stage breakdown per case	Low

7. User Flows

7.1 Case Identification → Case Creation Flow

```
Field Worker                  System                        Reviewer
  |                             |                               |
  ├── Enter ITS number ─────────►|                               |
  |                             ├── Lookup DB; if not found      |
  |                             ├── Fetch from ITS API           |
  |◄── Auto-fill applicant ─────|                               |
  |    (name, photo, contact)   |                               |
  |                             |                               |
  ├── Enter income/family data ►|                               |
  ├── Select "eligible_in" ─────►|                               |
  ├── Submit identification ────►|                               |
  |                             ├── Check for duplicate ITS      |
  |                             ├── Save identification record   |
  |◄── "Pending review" ────────|                               |
  |                             |                               |
  |                             |◄── Reviewer opens record ─────|
  |                             |◄── Marks as "eligible" ───────|
  |                             ├── Auto-create Case (BS-XXXX)   |
  |                             ├── Assign to DCM queue          |
  |                             |──────────────────────────────►|
```

7.2 Case Workflow — Counseling to Executive Approval

```
DCM                   Counselor             Welfare             ZI / Executive
  |                      |                     |                     |
  ├── Assign Counselor ──►|                     |                     |
  |                      |                     |                     |
  |                      ├── Fill Form ───────► |                     |
  |                      |   (7 sections)       |                     |
  |                      ├── Mark Complete ────►|                     |
  |                      |                     |                     |
  |◄─────────────────────|                     |                     |
  ├── Submit to Welfare ──────────────────────►|                     |
  |                                            |                     |
  |                                            ├── Review checklist    |
  |                                            ├── Welfare Approve ───►|
  |                                            |   (or reject → DCM)   |
  |                                            |                     |
  |                                            |                     ├── ZI Approve
  |                                            |                     ├── Executive L1 Approve
  |                                            |                     ├── Executive L2 Approve
  |                                            |                     └── → Finance Stage
```

7.3 Payment Schedule & Disbursement Flow

```
Finance                       System
  |                             |
  ├── Open Payment Schedule ────►|
  |                             ├── Load case financial data
  |◄── Show schedule form ──────|
  |                             |
  ├── Enter disbursement ───────►|
  |   (year, month, amount,     |
  |    repayment months)        |
  |                             ├── Generate repayment entries
  |◄── Schedule saved ──────────|
  |                             |
  ├── Confirm Disbursement ─────►|
  |   (actual_date)             ├── Mark is_disbursed = true
  |                             ├── Record disbursed_by + at
  |◄── Disbursement confirmed ──|
```

7.4 Fund Utilization Report Flow

```
Counselor / DCM                System                        Admin
  |                            |                              |
  ├── Open Fund Utilization ──►|                              |
  |                            ├── Load form config           |
  |                            |   (sections + questions)     |
  |◄── Dynamic form rendered ──|                              |
  |                            |                              |
  ├── Fill responses ──────────►|                             |
  ├── Upload photos ───────────►| (shop, stock, work, other)  |
  ├── Submit report ───────────►|                             |
  |                            ├── Save report + responses    |
  |◄── Report saved ───────────|                              |
  |                            |                              |
  ├── Request PDF ─────────────►|                             |
  |                            ├── Aggregate report data      |
  |                            ├── Build PDF via PDFKit/Puppet |
  |◄── PDF downloaded ─────────|                              |
```

8. Information Architecture

8.1 Navigation Structure by Role

Super Admin / Admin:

```
Dashboard
├── Case Identification
├── Cases
│   ├── All Cases
│   ├── Case Details (→ Counseling Form, Cover Letter, Payment, Fund Utilization)
│   └── Closure Management
├── Applicants
├── Users
│   └── User Management
├── Role Management
│   └── Roles & Permissions
├── Master Data
│   ├── Jamiat Management
│   ├── Jamaat Management
│   ├── Case Types
│   ├── Workflow Stages
│   ├── Executive Levels
│   ├── Relations / Education / Occupations
│   └── Fund Utilization Form Builder
├── Welfare Checklist Admin
├── Reports
├── Notifications
└── Profile
```

DCM / Counselor:

```
Dashboard
├── Case Identification
├── Cases (scoped to Jamiat/Jamaat)
│   └── Case Details (→ Counseling Form, Cover Letter, Payment, Fund Utilization)
├── Welfare Checklist
├── Reports
├── Notifications
└── Profile
```

Welfare Reviewer / ZI / Executive:

```
Dashboard
├── Cases (pending action at my stage)
│   └── Case Details (read + approval actions)
├── Welfare Checklist (review)
├── Notifications
└── Profile
```

Finance:

```
Dashboard
├── Cases (pending disbursement)
│   └── Payment Schedule Management
├── Fund Utilization (view)
├── Notifications
└── Profile
```

---

9. Technical Architecture Summary

9.1 Technology Stack

LAYER	TECHNOLOGY	VERSION / DETAIL
Backend	Node.js + Express.js	Node v16+ / Express v4.18
Frontend	React (functional components + hooks)	v18.2
Database	MySQL (InnoDB)	v8.0+
Authentication	JWT + bcryptjs	jsonwebtoken v9
State Management	React Query	v3.39
Forms	React Hook Form	v7.48
Styling	Tailwind CSS	v3.4
HTTP Client	Axios	v1.6
Routing	React Router	v6
Charts	Recharts	v2.8
File Upload	Multer	v1.4
PDF Generation	PDFKit + Puppeteer	v0.14 / v24
Excel	xlsx	v0.18
Validation	Joi	v17.11
Security	Helmet, CORS, express-rate-limit	Latest
Email	Nodemailer	v6.9 (optional)
Drag & Drop	@dnd-kit	Latest

9.2 Architecture Pattern

FACTS follows a **REST API + SPA** architecture. The Express backend exposes 150+ API endpoints organized by resource. The React frontend consumes these endpoints via Axios with React Query managing server-state caching. Authentication is stateless via JWT.

```
Browser (React SPA)
  ↕ HTTP/HTTPS (Axios + React Query)
Express API (Node.js)
  ↕ Parameterized Queries (mysql2)
MySQL Database (InnoDB)
  ↕ Multer
Filesystem (uploads/)
  ↕ External HTTP
ITS API (applicant data source)
```

9.3 Key Design Patterns

PATTERN	USAGE
REST Resource Controllers	25 route files; each resource is a self-contained Express router with CRUD + actions
JWT Middleware	Every protected route verifies token and loads user + permissions before handler runs
Scope Filtering	Middleware injects Jamiat/Jamaat scope into queries based on user's assigned role
Parameterized Queries	All DB queries use `mysql2` parameterized binding — no string concatenation
Soft Deletes	Workflow stages and roles use `is_active` flag; records are never hard-deleted

PATTERN	USAGE
Junction Tables	Many-to-many: user↔role, role↔workflow_stage, user↔workflow_stage
Audit Columns	`created_by`, `updated_by`, `created_at`, `updated_at` on all core tables

10. Data Model Overview

10.1 Core Entities

ENTITY	TABLE	PURPOSE
User	`users`	All user accounts with credentials and ITS reference
Role	`roles`	Role definitions with display name and permission JSON
User Role	`user_roles`	User-to-role assignments with Jamiat/Jamaat scope
Role Permission	`role_permissions`	Granular resource + action permission records per role
Case	`cases`	Core case records with status, workflow stage, and assignment
Applicant	`applicants`	Applicant profiles linked to cases
Case Identification	`case_identifications`	Pre-assessment eligibility records
Counseling Form	`counseling_forms`	Composite form record linking 7 section sub-tables
Cover Letter Form	`cover_letter_forms`	Cover letter data with multi-party signatures
Payment Schedule	`payment_schedules`	Yearly disbursement plans per case
Payment Repayment	`payment_schedule_repayments`	Monthly repayment entries per schedule
Fund Utilization Report	`fund_utilization_reports`	Quarterly reassessment reports per case
Workflow Stage	`workflow_stages`	Configurable approval stage definitions with SLA
Status History	`status_history`	Immutable log of every case status transition
Notification	`notifications`	In-app notification records per user
Case Attachment	`case_attachments`	File upload metadata linked to cases
Welfare Checklist	`welfare_checklist_items` / `_responses`	Configurable checklist items and per-case responses

10.2 Counseling Form Sub-Tables

SECTION	TABLE
Personal Details	`personal_details`
Family Details	`family_details` + `family_members`
Assessment	`assessment` + `assessment_products_services`
Financial Assistance	`financial_assistance` + `financial_assistance_timeline` + `financial_assistance_mentors` + `financial_assistance_action_plan`
Economic Growth	`economic_growth` + `economic_growth_projections`
Declaration	`declaration`
Attachments	`attachments`

10.3 Entity Relationship (Simplified)

```
Users ─── user_roles ─── Roles ─── role_permissions
  │
  ├── Cases ──────────── Applicants
  │     │
  │     ├── Counseling Forms (7 sub-tables)
  │     ├── Cover Letter Forms
  │     ├── Case Attachments
  │     ├── Status History
  │     ├── Case Comments + Workflow Comments
  │     ├── Payment Schedules ── Repayments
  │     ├── Fund Utilization Reports ── Photos + Files
  │     ├── Welfare Checklist Responses
  │     └── Case Closure ── Closure Documents
  │
  ├── Case Identifications ── Income Change Logs
  └── Notifications

Workflow Stages ── workflow_stage_roles ── Roles
               └── workflow_stage_users ── Users
```

---

11. Performance & Reliability

11.1 Performance Targets

OPERATION	TARGET TIME
Standard API response	< 500ms
Page load (React SPA)	< 3 seconds
Complex JOIN query (cases list)	< 2 seconds
PDF cover letter generation	< 5 seconds
Quarterly report PDF generation	< 10 seconds
Excel import (bulk users)	< 15 seconds
ITS API fetch (single)	< 3 seconds

11.2 Reliability Targets

REQUIREMENT	TARGET
System uptime	≥ 99.5% (excluding planned maintenance)
Data integrity	InnoDB foreign key constraints + parameterized queries on all operations
Error handling	Structured JSON error responses; no stack traces exposed to clients in production
Audit trail completeness	100% of status changes, income edits, and approval actions logged

11.3 Optimization Strategies

- MySQL connection pool (10 connections) via `mysql2` to avoid connection overhead
- React Query client-side caching with configurable stale times per resource
  Pagination with `LIMIT` / `OFFSET` on all list endpoints
- Specific column selection in queries; no `SELECT *` on production tables
- ITS API results cached in the local DB to avoid repeated external calls
- Multer file size limits enforced server-side before reaching business logic

---

12. Security & Compliance

12.1 Authentication Security

MEASURE	IMPLEMENTATION
Password hashing	bcryptjs — no plain-text passwords stored
JWT tokens	Signed with strong secret; 24-hour expiry; verified on every request
User-status check	Token verification also checks `is_active = 1` — deactivated users are blocked immediately
Login rate limiting	Configurable max attempts per IP window via `express-rate-limit`

12.2 Data Protection

THREAT	PROTECTION
SQL Injection	100% parameterized queries via `mysql2` — no string concatenation in SQL
XSS	React's JSX escapes output by default; API responses are JSON only
CSRF	Stateless JWT eliminates session cookie CSRF vectors
Unauthorized access	RBAC middleware on every route; scope filtering injects Jamiat/Jamaat constraints into DB queries
Input validation	Joi schema validation on all POST/PUT request bodies

12.3 File Security

MEASURE	IMPLEMENTATION
File type validation	Multer MIME-type + extension whitelist (PDF, DOC, DOCX, XLS, XLSX, JPG, PNG)
File size limit	10 MB per file enforced by Multer before file is written to disk
Isolated storage	Files stored in `uploads/` directory; never served directly — only via authenticated download endpoint

12.4 Infrastructure Security

MEASURE	IMPLEMENTATION
HTTP security headers	Helmet middleware sets Content-Security-Policy, X-Frame-Options, etc.
CORS	Restricted to configured frontend origin only
Environment secrets	DB credentials and JWT secret in `.env`; never committed to version control
HTTPS	Required in production via Nginx SSL termination

13. Deployment & Infrastructure

13.1 Production Environment

COMPONENT	SPECIFICATION
Runtime	Node.js v16+ with PM2 process manager
Database	MySQL 8.0+ (InnoDB engine)
Web Server / Proxy	Nginx (SSL termination, static file serving, API proxy)
Frontend	React production build served as static files via Nginx
File Storage	Local filesystem `uploads/` directory (S3 in roadmap)
OS	Ubuntu 22.04 LTS recommended

13.2 Deployment Architecture

```
Client Browser
    │ HTTPS
Nginx (port 443)
    ├── /           → React build (static)
    ├── /api        → Express API (port 5000, PM2)
    └── /uploads    → Express file serving
         │
    MySQL (local or managed DB)
    uploads/ (local filesystem)
    ITS External API (outbound)
```

13.3 Local Development

COMPONENT	SETUP
Backend	`cd backend && npm run dev` (nodemon on port 5000)
Frontend	`cd frontend && npm start` (React dev server on port 3000)
Database	Local MySQL; import `database/baaseteen_production_database.sql`
Combined	`npm run dev` from root (concurrently launches both)

---

14. Assumptions & Constraints

14.1 Assumptions

ID	ASSUMPTION
A-1	Users have access to modern web browsers (Chrome, Firefox, Safari, Edge)
A-2	The external ITS API is accessible from the server network at runtime
A-3	Each applicant has a unique ITS number that serves as their primary identifier
A-4	Organizational hierarchy follows a two-level structure: Jamiat (parent) → Jamaat (child)
A-5	Assessment scoring for Qardan Hasana follows fixed repayment structures
A-6	Concurrent users will not exceed 100 at any time in the current deployment
A-7	File storage will remain within server disk capacity; S3 migration planned for scale

14.2 Constraints

ID	CONSTRAINT
C-1	Technology stack is Node.js + React + MySQL — no framework migration planned
C-2	File storage is local filesystem only; no cloud object storage currently integrated
C-3	PDF generation requires PDFKit or Puppeteer; Puppeteer needs Chromium on the server
C-4	ITS API is an external dependency; outages degrade auto-fill functionality but do not block system use
C-5	Single-server deployment — no load balancing or horizontal scaling currently
C-6	Email notifications require SMTP configuration; optional and disabled by default
C-7	No real-time WebSocket support; notifications require page refresh or manual polling

---

15. Risks & Mitigation

ID	RISK	LIKELIHOOD	IMPACT	MITIGATION STRATEGY
R-1	Data loss / DB corruption	Low	Critical	Automated DB backups; InnoDB foreign keys; parameterized queries throughout
R-2	ITS API downtime	Medium	Medium	Graceful fallback to local DB; manual data entry remains available
R-3	Unauthorized cross-Jamiat data access	Low	High	Scope filtering enforced in middleware on every query; tested per role
R-4	Duplicate active cases per applicant	Medium	High	Server-side one-active-case-per-ITS enforcement before case creation
R-5	File upload abuse	Medium	Medium	Multer type/size validation; files stored outside web root; served via authenticated endpoint
R-6	Performance degradation at scale	Medium	Medium	DB indexing on all FKs and search fields; pagination; React Query caching
R-7	Puppeteer PDF failures	Low	Medium	PDFKit as alternative generator; error handling returns clear failure message
R-8	SLA misconfiguration	Medium	Low	Admin UI for SLA editing; SLA calculation isolated in `slaCalculator.js` utility

---

16. Out of Scope & Future Roadmap

16.1 Currently Out of Scope

ITEM	REASON
Native mobile app	Responsive web design serves mobile needs adequately
SMS notifications	Email + in-app notifications satisfy current requirements
Video conferencing	External tools used for live counseling sessions
Cloud file storage (S3)	Local filesystem is sufficient at current scale
Multi-tenant architecture	Single-organization deployment only
Public applicant portal	All intake is handled by field workers on behalf of applicants
Advanced BI dashboards	Simple analytics are sufficient; dedicated BI tools can connect to DB if needed

16.2 Future Roadmap Considerations

PRIORITY	FEATURE	DESCRIPTION
High	Email / SMS alerts	Send notifications via email or SMS for critical workflow events
High	Cloud file storage (S3)	Move uploaded files to S3 for scalability, redundancy, and CDN delivery
High	REST API public documentation	OpenAPI/Swagger documentation for third-party integrations
Medium	Advanced analytics	Charts, trends, and KPI tracking for case volumes, SLA performance, and disbursements
Medium	Self-service password reset	Email-based password reset without admin intervention
Medium	Applicant self-service portal	Allow applicants to view their case status via a secure read-only portal
Low	WebSocket notifications	Replace polling with real-time push via WebSocket
Low	Native mobile app	React Native app for field workers to perform case identification offline
Low	Multi-language support	Interface translations for Urdu and other languages

17. Glossary

TERM	DEFINITION
FACTS	File and Case Tracking System — this platform
PRD	Product Requirements Document — this document
Baaseteen	Primary welfare organization operating this system
SHND	Social Help and Development — secondary organization served by FACTS
ITS	Institutional Information System — external system providing applicant identity data
ITS Number	Unique 8-digit identifier assigned to each community member by the ITS system
Case	A welfare application linked to a single applicant, progressing through workflow stages
Case Identification	Pre-assessment eligibility screening record created before a full case is opened
Jamiat	Top-level organizational unit (zone/region)
Jamaat	Sub-unit under a Jamiat (local congregation/group)
DCM	Deputy Case Manager — staff member responsible for creating and managing cases
ZI	Zonal Incharge — approver at the zonal level
Enayat	Grant-type financial assistance (non-repayable)
Qardan Hasana (QH)	Interest-free loan-type financial assistance (repayable)
Manzoori	Sanction — the final approval stage before financial disbursement
RBAC	Role-Based Access Control — restricting system access based on user roles
SLA	Service Level Agreement — maximum time a case should remain at a workflow stage
JWT	JSON Web Token — stateless authentication token issued at login
Fund Utilization	Quarterly tracking of how disbursed financial assistance is being used by the applicant
Counseling Form	Multi-section digital form documenting the full welfare counseling assessment
Cover Letter	Summary document generated for executive review of the case

— End of Product Requirements Document —

FACTS — File & Case Tracking System
Version 3.7 — April 23, 2025